



Lab 03: DNS & DHCP Servers

DNS · DHCP · Google · Facebook Servers · Switch · PC0 · PC1 · PC2

Lab #03

Topic: Network Services

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure a DHCP server to automatically assign IPs to PC0, PC1 and PC2.
- Configure a DNS server to resolve hostnames (google.lab, facebook.lab) to IPs.
- Set up simulated 'Google' and 'Facebook' servers with static IPs.
- Verify DNS resolution and DHCP lease acquisition from all three PCs.
- Relate to Network+ objective 1.6: DNS and DHCP services.

2. Network Topology

A central 2960 switch (Switch0) connects seven devices: DNS server (Fa0/5), DHCP server (Fa0/1), Google server (Fa0/6), Facebook server (Fa0/7), PC0 (Fa0/2), PC1 (Fa0/3) and PC2 (Fa0/4). All devices are on the same subnet.

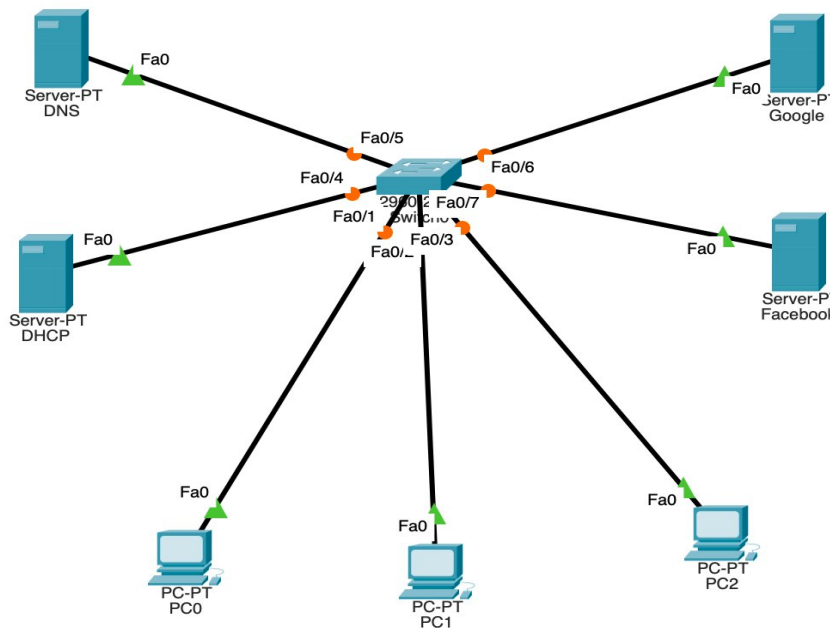


Figure 1 – Star topology: DNS, DHCP, Google, Facebook servers and 3 client PCs on Switch0

Device	Role	Port on Switch	IP Address	Notes
DNS	DNS Server	Fa0/5	192.168.1.2	Static — resolves hostnames
DHCP	DHCP Server	Fa0/1	192.168.1.3	Static — assigns IPs to clients



Google	Web Server	Fa0/6	192.168.1.10	Simulated google.lab
Facebook	Web Server	Fa0/7	192.168.1.20	Simulated facebook.lab
PC0	Client	Fa0/2	DHCP	Receives IP from DHCP
PC1	Client	Fa0/3	DHCP	Receives IP from DHCP
PC2	Client	Fa0/4	DHCP	Receives IP from DHCP

3. Step-by-Step Configuration

3.1 DHCP Server Configuration

Click DHCP server → Services → DHCP → turn On. Create pool:

Pool Name	LAN_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP	192.168.1.50
Subnet Mask	255.255.255.0
Max Users	50

3.2 DNS Server Configuration

Click DNS server → Services → DNS → turn On. Add A records:

Hostname	Record Type	Address
google.lab	A Record	192.168.1.10
facebook.lab	A Record	192.168.1.20
dhcp.lab	A Record	192.168.1.3

3.3 Set PCs to DHCP

PC0/PC1/PC2 > Desktop > IP Configuration > select DHCP

Wait 2-3 seconds – IP auto-assigned from pool (192.168.1.50+)

DNS server field auto-filled: 192.168.1.2

Default Gateway auto-filled: 192.168.1.1

4. Verification

PC0> ipconfig ! Confirm DHCP lease

PC0> ping 192.168.1.2 ! Ping DNS server

PC0> ping google.lab ! Test DNS name resolution

PC0> ping facebook.lab ! Test second DNS record



5. Key Takeaways

- DHCP uses the 4-step DORA process: Discover, Offer, Request, Acknowledge.
- DNS A records map hostnames to IPv4 addresses — the most common record type.
- DHCP option 6 (DNS server) and option 3 (router/gateway) are distributed with the lease.
- Simulating real servers (Google, Facebook) helps practice DNS resolution in a safe lab.
- Always assign static IPs to servers — DHCP is for client devices only in production.